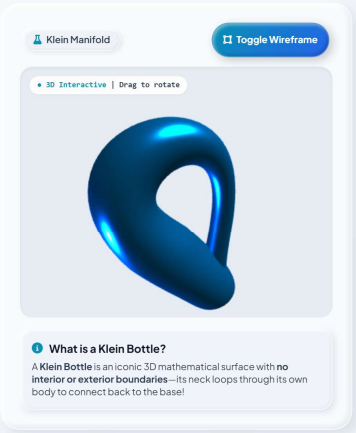
B.E. Mechanical Engineering (2023–2027)

Hi, I'm VASANTHKUMAR N

Mechanical Engineer specializing in CAD Modeling, ANSYS FEA, and IIoT Digital Twins.

Combining 3D design, finite element analysis, sensor telemetry, and machine learning to build intelligent mechanical products and predictive maintenance frameworks.

vasanth19102005@gmail.com | +91 6369797334 | Madurai, Tamil Nadu



ENGINEERING PHILOSOPHY

Bridging Physical CAD & Digital Intelligence.

I am a B.E. Mechanical Engineering student at Kamaraj College of Engineering & Technology (Class of 2027). My work converges classical 3D CAD modeling and ANSYS FEA simulation with modern Industry 4.0 IoT telemetry and machine learning.

4+ Projects — CAD & IIoT	3+ Core Domains — CAD, FEA & IIoT
6+ Certifications — NPTEL Elite, Creo	2027 Graduation — KCET College

01. PRECISION DESIGN

3D CAD & FEA Simulation

Modeling parametric assemblies in Creo, Fusion 360 & CATIA V5, validated with ANSYS thermal and static structural stress analysis.

02. SMART TELEMETRY

IIoT & ML Predictive Analytics

Integrating ESP8266 microcontrollers, ADXL345 vibration sensors, and Random Forest ML models to detect real-time machinery faults.

03. DIGITAL TWINS

3D Unity & MSME Innovation

Developing real-time synchronized 3D Unity dashboards to bring affordable Industry 4.0 predictive maintenance to manufacturing MSMEs.

TECHNICAL EXPERTISE

Engineering & Software Skills

Comprehensive toolkit spanning 3D CAD modeling, FEA simulation, industrial IoT telemetry, and modern manufacturing.

CAD Software

3D parametric modeling, assembly design, surface modeling, and 2D drafting.

AutoCAD (2D Drafting)	Advanced
PTC Creo (3D CAD)	Certified
Autodesk Fusion 360	Advanced
CATIA V5	Certified

Simulation & FEA

Structural, thermal, dynamic, and modal boundary condition simulation.

ANSYS Static Structural	Proficient
ANSYS Steady / Thermal	Proficient
ANSYS Modal Analysis	Proficient

IIoT & Digital Twins

Sensor hardware telemetry, cloud databases, ML prediction, and 3D visualization.

ESP8266 Microcontrollers	Hands-on
ADXL345 & DS18B20	Telemetry
Random Forest ML	Python
Firebase & Unity 3D	Integration

Manufacturing & QC

Production planning, CNC G-code programming, and geometric dimensioning.

CNC Programming	NTTF Certified
GD&T Tolerancing	Applied
Metrology & Inspection	Applied

3D Rendering & Media

Product rendering, animation, asset modeling, and video production.

Blender (3D Render)	Intermediate
DaVinci Resolve	Intermediate

Core Strengths

Problem solving, collaborative design, and product lifecycle development.

- ✓ Problem Solving
- ✓ Creative Thinking
- ✓ Team Collaboration
- ✓ 3D Part Modeling

PORTFOLIO SHOWCASE

Engineering & IIoT Projects

Explore key projects showcasing mechanical design, FEA validation, IIoT telemetry, and physical prototyping.

All | CAD & FEA | IIoT & Twins | Industrial

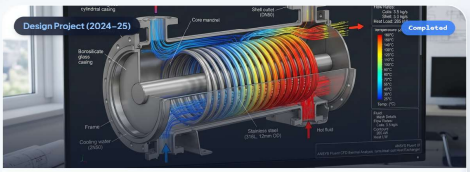


Digital Twin-Based Predictive Maintenance System

Developed a Digital Twin framework integrating ANSYS FEA, ESP8266 IoT, Firebase cloud database, Unity 3D visualization, and Machine Learning for real-time health monitoring of rotating machinery.

ANSYS FEA | ESP8266 | Firebase | Unity 3D | Random Forest ML

MSME Smart Manufacturing — View Details >



Design & Fabrication of Compact Heat Recovery System

Designed a shell-and-helical-coil heat exchanger in Fusion 360, performed thermal and structural stress analysis in ANSYS Workbench, and fabricated a working physical prototype to measure heat transfer effectiveness.

Fusion 360 | ANSYS Workbench | Thermal Analysis | Fabrication

Waste Heat Recovery — View Details >



IoT / RFID Sewing Performance Tracking System

Implemented an RFID shop-floor tracking system with ESP8266 modules and a live central dashboard to monitor operator sewing output, eliminate data-entry errors, and resolve bottlenecks in real time.

ESP8266 | RFID Hardware | Live Dashboard

Shop Floor Automation — View Details >



Smart Non-Newtonian Fluid Dynamic Speed Bump

Engineered an adaptive speed breaker utilizing shear-thickening non-Newtonian fluid (STF) that yields smoothly under low-speed vehicle roll while instantaneously solidifying into a rigid obstacle upon high-speed impact.

Non-Newtonian Fluid | Shear-Thickening (STF) | Rheology Modeling | Elastomer Housing

Smart Traffic Rheology — View Details >

CAREER HISTORY

Internships & Industrial Exposure

Hands-on experience in manufacturing paint shop operations and industrial IoT integration.

INDUSTRIAL PROJECT

Pentagon Apparels Unit II, Virudhunagar

IoT / RFID Sewing Performance Tracking

Implemented real-time operator-level output tracking using ESP8266 modules and RFID card readers. Integrated live web dashboards for supervisors to detect bottlenecks instantly.

PAINT SHOP INTERNSHIP (15 DAYS)

Schwing Stetter India Pvt Ltd

Paint Shop Operations

Studied end-to-end industrial surface preparation, coating application, and curing stages. Observed quality control checkpoints for finish consistency, defect identification, and industrial safety compliance.

CREDENTIALS

Certifications & Achievements

Validated technical competencies, symposium awards, and professional memberships.

Professional Certifications

- **NPTEL Elite Certificate – Product Design & Manufacturing**
  IIT Kanpur / SWAYAM
- **Industrial Internet of Things (IIoT) Training**
  Hands-on Training at SASTRA University, Thanjavur
- **CATIA V5 Certification**
  Kamaraj College of Engineering & Technology
- **Advanced Creo (3D CAD)**
  Udemy Professional Certificate
- **CNC Programming Training**
  NTTF (Nettur Technical Training Foundation), Bangalore

Awards & Memberships

- **2nd Prize – Technical Quiz**
  Yuvasakthi National Technical Symposium, NPR College
- **3rd Prize – SkyForge Event**
  Euphoria Techno Management Meet, Kalasalingam Academy
- **2nd Prize – SkyForge Event**
  Euphoria Techno Management Meet, Kalasalingam Academy (2026)
- **ISTE Student Member**
  Indian Society for Technical Education (SM No. 11709, 2025–2027)

LET'S CONNECT

Get In Touch

Actively seeking early-career opportunities in **Design Engineering**, **Manufacturing Engineering**, and **Industrial IoT / Smart Manufacturing**. Open to full-time roles, internships, and technical collaborations—feel free to reach out directly or connect!

Email Address
vasanth19102005@gmail.com

Phone / WhatsApp
+91 6369797334

LinkedIn Profile
vasanthkumarn1910

Send a Direct Message

WhatsApp Quick Chat

Your Name: John Doe

Your Email: john@example.com

Subject: Job Opportunity / Project Inquiry

Message: Hi Vasanthkumar, I would like to discuss...

Send Message